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EXPERIMENT-4

Basic Output Stage Topologies

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1 Introduction

1.1 Class A Follower

[1]The Class A output stage represents the output stage of an amplifier circuit and features transistors that operate continuously, ensuring accuracy and low distortion over the entire wave period of the signal.

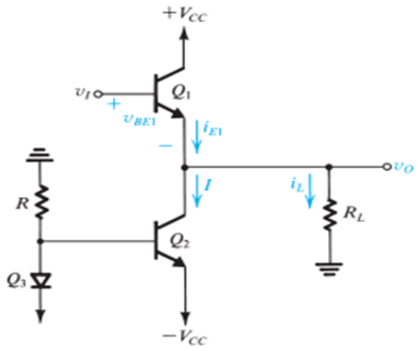


Figure 1. An emitter follower (Q1) biased with a constant current I supplied by transistor Q2.

1.2 Class B Follower

Class B output stage refers to the output stage of an amplifier circuit and is usually designed for energy efficiency. In this type of amplifier, the output transistors operate only on the positive half-wave or negative half-wave of the signal, so that the transistors remain in the off state most of the time.

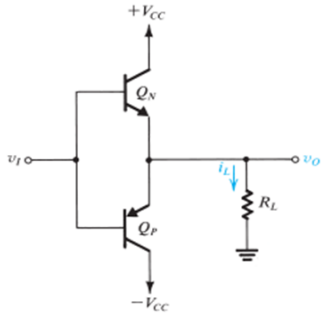


Figure 2. A class B output stage.

1.3 Diode Biased Class-AB Output Stage

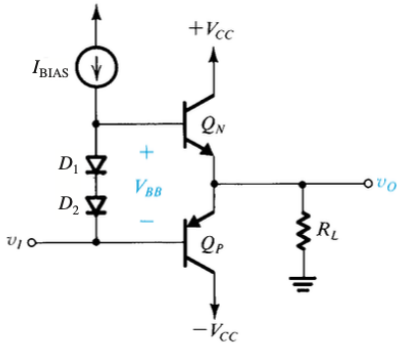


Figure 3. A class AB output stage biased by diodes.

Crossover distortion in Class B output stage can be eliminated by biasing the output transistors by a small current. This circuit is called AB output stage amplifier. In circuit Figure 3 an AB circuit which the bias voltage V_{bb} is generated by passing a constant current I_{bias} by trough a pair of diodes.

2 Brief Theory and Theoretical Calculations

2.1 Class A Follower

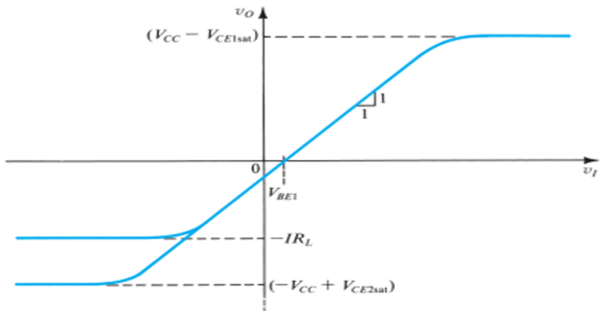


Figure 4. Transfer characteristic of the emitter follower in Fig. 1.

According to the Figure 4. graph, in the Class A Output Stage circuit, the upper limit is determined by V_{cc}-V_{CE1sat}, while the lower limit varies according to the RL value.

2.2 Class B Follower

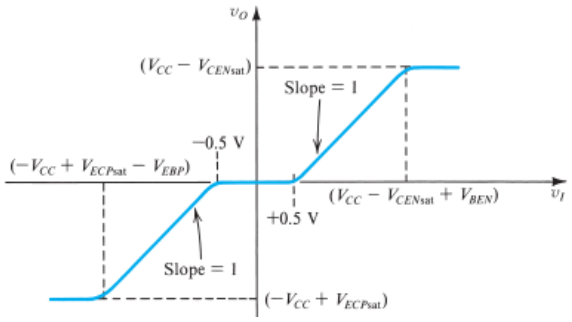


Figure 5.a. Transfer characteristic of the class B output stage amplifier.

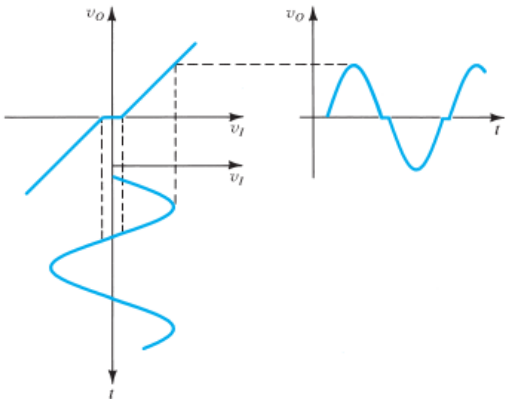


Figure 5.b. Illustrating how the dead band in the class B transfer characteristic results in crossover distortion.

[1]As shown in Figure 5.b, the dead band condition in the class B output stage transfer characteristic graph causes crossover distortion.

2.3 Diode Biased Class-AB Output Stage

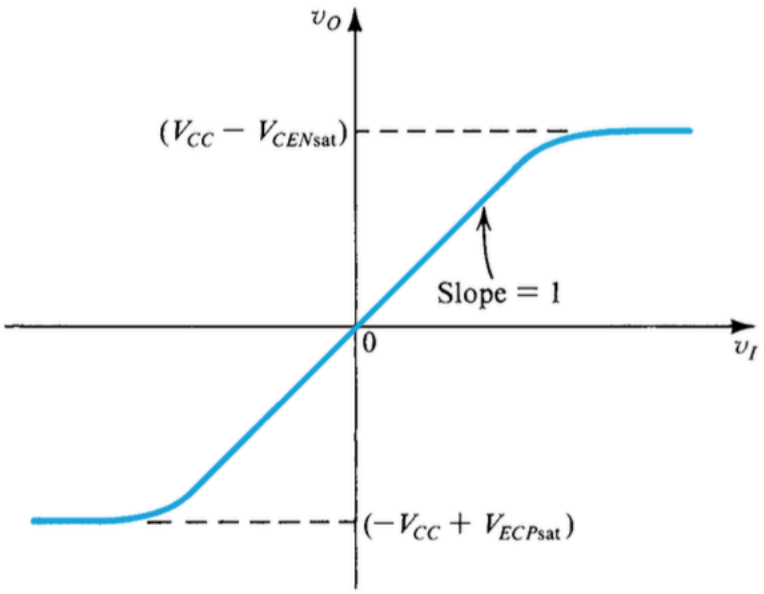


Figure 6. Transfer characteristic of the class AB output stage amplifier.

As shown in Figure 6, for Diode Biased Class-AB Output Stage amplifier the dead band of the class B amplifier does not appear, and the input signal is transferred to the output without distortion.

3 Experimental Results (Figures, tables, graphs)

3.1 Class A Follower

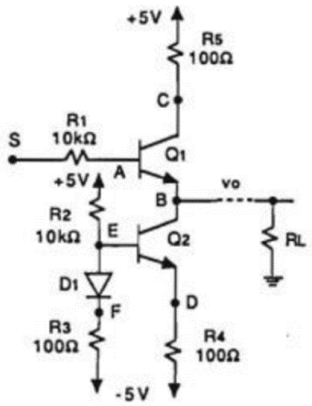


Figure 7. Class A output stage in experiment.

Table 1: DC values of Class-A amplifier

	V_A	V_B	V_C	V_D	V_E	V_F	I_C	I_B	β
a)	-31mV	-0,64V	4,95V	-4,9 V	-4,27V	-4,94V	0,5 mA	3,1 μA	161,3
b)	-0,32V	-1,07V	4,01V	-3,8V	-3,27V	-4 V	3,9mA	32 μA	30

Firstly, the circuit shown in Figure 7. was established, node S was grounded and the first measurements in table 1 were made with R2=10 kΩ. Then the measurements were repeated by shunting a 1 kΩ resistor to R2. As a result, due to the change in RL, it is seen that B, which affects many factors such as gain, changes.

Table 2: AC values of Class-A Amplifier

	v_s	v_a	v_b	v_b/v_s	v_b/v_a	R_{in}
$R_L = 10k\Omega$	1,06mV	1,02 mV	1,10 mV	1,03	1,07	1,63MΩ
$R_L = 1k\Omega$	104mV	98 mV	104mV	1	1,06	150kΩ

The circuit was connected so that R2 =10kΩ, RL =10kΩ. Node S was connected to a generator providing a triangular wave of 0.1 Vpp at 1kHz. The peak-to-peak voltages of nodes S, A, B were measured. The voltage gains from S to B and from A to B and the input resistance at A were calculated. These processes were repeated for RL=1 kΩ.

$$Rin = \frac{Vs - Vb}{Ib} \left(for\ RL\ = 10k\Omega => Rin = (106 - 102) * \frac{103}{3.1 * 10^{-6}} = 1.3\ M\Omega \right)$$
$$\left(for\ RL\ = 1k\Omega => Rin = (104 - 98) * \frac{103}{32*10-6} = 188\ k\Omega \right)$$

Table 3: Critical Peak Voltages of Class-A amplifier

	v_s when v_B^+	v_s when v_B^-
$R_L = 10k\Omega$	12V	10 V
$R_L = 1k\Omega$	12V	1,20V

Then, for RL=10 kΩ, RL=1 kΩ cases, S and B nodes were observed on the oscilloscope and the limit values of B were tried to be obtained by increasing and decreasing the voltage of S and noted in Table 3. When the values are analyzed, it is seen that the change in RL does not affect the upper limit, but the decrease in RL decreases the value of the lower limit.

3.2 Class B Follower

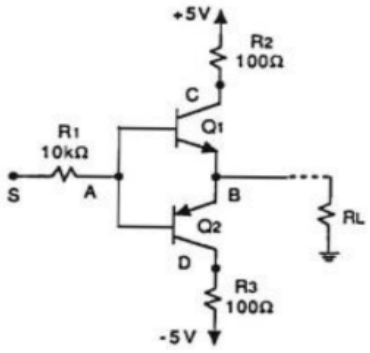


Figure 8. A class B output stage in experiment.

Table 4: DC values of Class-B amplifier

V_A	V_B	V_C	V_D
0,2mV	0,8mV	5,03V	-5,06V

The circuit was established as shown in Figure 8, node S was grounded, RL=10kohm was selected and nodes A,B,C,D were measured with a multimeter. The amount of DC current in this circuit is very small because the main purpose of this circuit is to increase the amplifier efficiency and as the amplifier efficiency increases, the amount of energy taken from power supplies is tried to be minimized.

Table 5: AC values of Class-B Amplifier

v_s	v_a	v_B	v_S^+	v_A^+	v_B^+	v_S^-	v_A^-	v_B^-
204mV	205mV	5mV	12,1V	11,4V	10,2V	12,1V	11,4V	10,2V

The circuit was connected so that RL = 10kΩ. Node S was connected to a generator providing a triangular wave of 0.2 Vpp at 1kHz. The peak-to-peak voltages of nodes S, A, B were measured. Then node B was observed to find the limit values of B. When node B was at 10.2 V, its upper and lower limits were reached. In this circuit, S and A values were measured and noted in Table 5.

3.3 Diode Biased Class-AB Output Stage

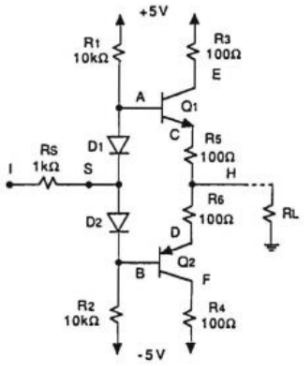


Figure 9. Class-AB Amplifier experiment circuit.

Table 6: DC values of Class-AB Amplifier

V_S	V_A	V_B	V_C	V_D	V_E	V_F	V_H	I_1	I_2
8mV	0,58V	-0,56V	24mV	-22mV	5,01V	-5,05V	7mV	0,2mA	0,17mA

Firstly, the circuit is setup as shown in Figure 9. To measure the dc operation of the amplifier. The node I is grounded, and no load connected. Then the measurements in Table 6 is taken.

Table 7: AC values of Class-AB Amplifier

v_s	v_i	v_h	v_i^+	v_h^+	v_i^-	v_h^-
0,170V	0,202V	0,114V	12,9V	8,96V	12,9V	8,96V

Than to circuit in Figure 9 a triangle wave generator with 0.2 Vpp at 1Khz is connected to node I, and a 10 Kohm load is connected. Then the voltages at nodes S, I and H measured. After that, the input voltage is raised to find the largest unclipped output available. When

the output voltage is clipped from the top side and from the bottom side the node voltage of H and I node is measured and noted.

4 Discussion

As a result, the working principles of Class A, Class B and Class A-B Output Stage circuits were learnt practically.

4.1 Class A Follower

While investigating the Class A output stage circuit, it was found that this circuit has low distortion and high-power consumption. Although it is bad in terms of energy efficiency, it provides an advantage in terms of accuracy. In the circuit shown in Figure 7, firstly, the effect of R2 change on B value was observed. Then, it was seen that the decrease in the RL value decreases the Rin value. In the same circuit, when the limit values of node B were observed according to the RL decrease, it was seen that the upper limit did not change while the lower limit of node B decreased.

4.2 Class B Follower

While researching the Class B output stage circuit, it was found that this circuit has high efficiency. In the circuit shown in Figure 8, the node B (output) was measured, and the distortion limits of B were observed. The upper and lower distortion limits were found to be the same and nodes A, B were measured.

4.3 Diode Biased Class-AB Output Stage

To make an amplifier with low distortion like Class A and high efficiency output of the class B, the Class AB amplifier is used [2]. In Class A amplifier the single output transistor conducts for the full 360° of the cycle input waveform. In class B the two output transistors only conduct for one half (180°) of the input wave form. In Class AB the two output transistors conduct somewhere between 180° and 360° of the input waveform. This causes a low power, and make the class AB efficiently than class A. But it is slightly less efficient than that Class B because the small current need to bias the transistors. The Class AB amplifier is used in various applications, like audio frequency, Tv receivers, CD players, etc.. [3].

5 References

[1] Smith, K. C. and Sedra, A. S., Laboratory Explorations for Microelectronic Circuits, Oxford University Press, 1998.
[2] https://www.electronics-tutorials.ws/amplifier/class-ab-amplifier.html
[3] https://www.elprocus.com/class-ab-amplifier/